

Digital twins, synthetic patient data, and in-silico trials: can they empower paediatric clinical trials?

Mohan Pammi, Prakesh S Shah, Liu K Yang, Joseph Hagan, Nima Aghaeepour, Josef Neu



Lancet Digit Health 2025;
7: 100851

Published Online May 13, 2025
<https://doi.org/10.1016/j.landig.2025.01.007>

Department of Pediatrics, Baylor College of Medicine and Texas Children's Hospital, Houston, TX, USA (Prof M Pammi MD, J Hagan ScD); Department of Paediatrics, Mount Sinai Hospital, University of Toronto, Toronto, ON, Canada (P S Shah MD); Stanford University School of Medicine, San Francisco, CA, USA (L K Yang PhD, N Aghaeepour PhD); University of Florida, Gainesville, FL, USA (Prof J Neu MD)

Correspondence to:
Prof Mohan Pammi, Department of Pediatrics, Baylor College of Medicine and Texas Children's Hospital, Houston, TX 77030, USA
mohanv@bcm.edu

Randomised controlled trials are the gold standard to assess the effectiveness and safety of clinical interventions; however, many paediatric trials are discontinued early due to challenges in patient enrolment. Hence, most paediatric clinical trials suffer from lack of adequate power. Additionally, trials are expensive and might expose patients to unproven therapies. Alternatives to overcome these issues using virtual patient data—namely, digital twins, synthetic patient data, and in-silico trials—are now possible due to rapid advances in digital health-care tools and interventions. However, such digital innovations have been rarely used in paediatric trials. In this Viewpoint, we propose using virtual patient data to empower paediatric trials. The use of virtual patient data has the advantages of decreased exposure of children to potentially ineffective or risky interventions, shorter trial durations leading to more rapid ascertainment of safety and effectiveness of interventions, and faster drug approvals. Use of virtual patient data could lead to more personalised treatment options with low costs and could result in faster clinical implementation of interventions in children. However, ethical and regulatory concerns, including replacing humans with digital data, data privacy, and security should be addressed and the safety and sustainability of digital data innovation ensured before virtual patient data are adopted widely.

Introduction

A well-designed, adequately powered, and diligently performed randomised controlled trial (RCT) is considered as high-certainty evidence for effectiveness of a clinical intervention for the specified outcome(s). However, RCTs are expensive and time-consuming, expose patients to unproven therapies until results are evident, and are hard to perform for heterogeneous chronic or rare diseases. Many trials are discontinued due to challenges in patient enrolment and retention. In addition, compared with RCTs in adults, RCTs in children, including neonates and preterm infants, have small sample sizes and lack adequate power.^{1,2} Therefore, intervention effects are frequently extrapolated from other populations to children and neonates or assumed by consensus from experts.

An example of a neonatal outcome that is often studied with underpowered trials is the outcome of necrotising enterocolitis, which has an incidence of 5–10% in preterm infants, born at less than 1500 g (low event rate).³ Interventions targeting necrotising enterocolitis necessitate large multicentre trials for adequate power, which require substantial resources including money and effort. Probiotics have the biological plausibility to prevent necrotising enterocolitis and have been investigated in clinical trials. Sharif and colleagues, in a systematic review and meta-analysis of probiotics in preterm infants, included 56 trials (10 812 infants).⁴ The median sample size of the included trials was dismally small at 149. Also, 21 of these trials enrolled fewer than 100 participants, 20 enrolled between 100 and 199 participants, 12 enrolled between 200 and 499 participants, and only three enrolled more than 500 participants. Sample size calculations for interventions for a modest (15–20%) change in the outcome exceed thousands⁵ and will take many years to complete at the expense of resources. None of the trials in the probiotics systematic review were close to the

numbers needed for adequate power (only a couple of trials had a good sample size, with the largest trial including 1315 participants). The majority of Cochrane reviews in the neonatal-perinatal field have concluded that further trials are needed due to low to very low certainty of evidence.

Alternatives to multicentre RCTs

What are the alternatives to large multicentre RCTs for an outcome with a low event rate similar to that of necrotising enterocolitis? Meta-analyses of RCTs that provide quantitative data synthesis of clinical trials increase precision and the ability to detect significant differences in the outcomes. However, clinical variability (heterogeneity) associated with the population, differences in intervention (dosage and duration or frequency), and variation in comparator groups (placebo or no treatment or other intervention) are major issues in combining and interpreting results from various studies.⁴ A planned prospective meta-analysis of trials by collaborators who perform similar individual trials is another option—eg, the Neonatal Oxygenation Prospective Meta-analysis (NeOProm) Collaboration on saturation targeting in preterm infants.⁶ However, a prospective meta-analysis that requires international collaboration and intent cannot always be planned. Tarnow-Mordi and collaborators have proposed large trials (megatrials) that are clinically robust and simple to conduct² and the Alpha international collaboration is trying to perform such megatrials;⁷ however, identifying a funding source and collaboration for such megatrials will remain a challenging endeavour. Advances in harnessing artificial intelligence (AI) in health care provide alternative solutions in testing clinical interventions using virtual patient data—namely, digital twins, synthetic patient data, and in-silico trials.

Digital twins

The National Academies of Sciences, Engineering, and Medicine define a digital twin as “a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system of systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value. The bidirectional interaction between the virtual and the physical is central to the digital twin.”⁸ This global definition of a digital twin is too broad to use in clinical care or in clinical trials in patients. Drummond and Gonsard define a patient-specific digital twin (patient digital twin) as “a viewable digital replica of a patient, organ, or biological system that contains multidimensional, patient-specific information”.⁹ The advantage of this definition is that it focuses on the patient and excludes biological systems (eg, cells and tissues) and cyber-physical systems (eg, cardiac defibrillators), which do not represent the patient. Digital twin technology uses AI-based cyber-physical systems and closed-loop optimisation using real-time patient data and can optimise treatment and research.¹⁰ A digital twin can provide synthetic data for the control arm of clinical trials and bolster sample size, which is much more cost-effective than traditional RCTs. Digital twins are different from external control arms that are incorporated into clinical trials.¹¹ External controls are selected from external data sources such as historical clinical trials, and real-world data such as electronic health records or registries. Digital twins use individual patients’ model-based estimates of outcomes in the absence of intervention for the control group.¹¹

As a significant accelerator of clinical trials, digital twins bring numerous advantages to modern medical care and drug discovery that facilitate their transformation from a reactive and remedial system to a proactive and integrated science aiming to provide patients with personalised interventions and disease management.^{12,13} Compared with the relatively well-studied digital twin system in the adult population, the digital twin system in paediatric health care has been inadequately studied but demonstrates great potential.¹⁴ Digital twins can be especially applicable to studies in preterm infants, in which true control groups are difficult to find because of the heterogeneity of the population. For example, a preterm infant born at 23 weeks of gestation has different physiological characteristics compared to a preterm infant born at 32 weeks of gestation, although both fall under the category of very preterm infants. Digital twins capture the health patterns of real children and serve as mirror objects that fully represent patients in virtual simulations, predictive analyses, and scenario testing. Children are unable to communicate responses to clinicians’ interventions as adults can. Therefore, their digital twins serve as representations to aid clinicians in gaining comprehensive understanding of their conditions and reactions to different treatments. A digital twin acts as a prototype for generating synthetic medical records that simulate patient conditions without corresponding to real-world individuals. This capability is

especially valuable for broadening the scope and diversity of data available for research on preterm neonates and rare diseases, which are infrequently encountered in the general population. Digital twins can help to mitigate bias and enhance the generalisability of research findings.

Drummond and Gonsard categorise patient-specific digital twins⁹ into simulated patient digital twins based on one-time assessments—personalised, viewable digital replicas of patients’ anatomy and physiology based on computational modelling to run simulations for predicting outcomes in hypothetical scenarios or evaluating therapeutic approaches; and monitoring patient digital twins focused on continuous tracking—personalised, viewable digital replicas of patients leveraging aggregated health data and analytics to enable continuous predictions of risks and outcomes over time and provide feedback for optimising care.

Despite being in its early stages, there has been some pioneering work in paediatric digital twins in recent years. Gonsard and colleagues reported a survey of 104 children (aged 8–17 years) with asthma on a potential digital twin system that integrates continuous data collection from children’s homes with AI-based recommendations to dynamically adjust children’s asthma care in real time.¹⁵ Gonsard and colleagues evaluated the readiness of children and adolescents to adopt the digital twin system for daily management of asthma through interviews. More than half (56%) were willing to use a digital twin system for the daily management of their asthma if it was as effective as current care, and up to 93% if it was more effective.¹⁵ In 2024, Sizemore and colleagues used generative AI to create a digital twin of the microbiome system, capturing dynamic bacterial interactions using faecal sample data from neonatal intensive care unit preterm infants.¹⁶ This approach provides a dynamic view of bacterial abundances in preterm infants, as their microbiomes continually evolve. An AI tool, Q-net, was then designed to simulate and predict changes in the infant microbiome.¹⁶ Inspired by large language models like GPT, Wang and colleagues introduced TWIN-GPT, a method that fine-tunes pre-trained GPT models with a focus on clinical trial datasets to create personalised digital twins.¹⁷ TWIN-GPT processes electronic health records to simulate patient-specific medical scenarios and predict potential medical events using structured prompt tasks and encoded data inputs.¹⁷ Although these efforts might not directly relate to clinical trials, they pave the way for the use of virtual patient data in paediatric clinical trials in the near future. Global centres specialising in and adopting digital twin technology for clinical research are growing exponentially and examples include the Swedish Digital Twin Consortium, Human Digital Twin OnePlanet Research Center, Empa research centre, DIGIPREDICT consortium, Living Heart Project, and COVID-19 Longhailer Advocacy Project.

Synthetic patient data

Advancement in generative AI models enables the development of AI-generated synthetic patient data.^{18,19}

For more on the **Swedish Digital Twin Consortium** see <https://www.sdttc.se/>

For more on the **Human Digital Twin OnePlanet Research Center** see <https://www.oneplanetresearch.com/innovation/human-digital-twin/>

For more on the **Empa research centre** see <https://www.empa.ch/web/s604/digital-twin>

For more on the **DIGIPREDICT consortium** see <https://www.digipredict.eu/>

For more on the **Living Heart Project** see <https://www.3ds.com/products-services/simulia/solutions/life-sciences-healthcare/the-living-heart-project/>

For more on the **COVID-19 Longhailer Advocacy Project** see <https://www.longhailer-advocacy.org/>

A universally accepted definition of synthetic data does not exist, which creates some problems in reproducibility and transparency in clinical research.²⁰ A working definition proposed by the Royal Society and The Alan Turing Institute states that synthetic data are “data that has been generated using a purpose-built mathematical model or algorithm, with the aim of solving (a set of) data science task(s)”.²¹ Synthetic data are computer-generated by modelling from existing datasets and mimic real-world patterns. Synthetic data can be used to support efficient medical and health-care research, while minimising the need to access personal data.^{18,22} Generative machine learning models may be first trained on synthetic datasets and allowed to learn the essential characteristics before being applied to the real patient dataset, thereby decreasing data privacy concerns and saving time. To the best of our knowledge, there are no examples of the use of synthetic data in paediatric clinical research. In an adult study of Alzheimer’s disease, investigators used data from the placebo arms of previous RCTs and observational studies (an existing dataset of 6919 patients including 64 clinical variables) and generative machine learning models to generate synthetic clinical records to forecast disease outcomes with reasonable accuracy.²³ In rare diseases in which the number of patients with the condition is small or in cases of incomplete data, generative synthetic data can supplement real-world evidence. Generative adversarial networks have been used to create synthetic data for a cohort of patients older than 18 years with myelodysplasia, a rare disease, which contained information on clinical features, genomics, treatment, and outcomes.²⁴ Synthetic data in this study mimicked real patient clinical-genomic features and outcomes.²⁴ Synthetic data derived from generative models could be a way to address low sample sizes in RCTs, thereby providing a less expensive way to obtain expedited results and implementation. However, synthetic data should have sufficient fidelity (quality of the data), represent the diversity of the real population, and be generalisable to the paediatric population or patient groups that are studied. Assessing biases in the source dataset for generative synthetic data pre-emptively might address some of the challenges.¹⁸

In-silico trials

A clinical trial that is conducted digitally through simulation and modelling is referred to as an in-silico trial,¹¹ which bypasses the problems of patient recruitment and might support both the control and intervention arms of a trial.²⁵ The Virtual Imaging Clinical Trial for Regulatory Evaluation study used computer-simulated images of virtual patients and compared the performance of digital mammography and digital breast tomosynthesis for detecting breast lesions (n=2986 virtual patients).²⁶ When the results of the in-silico trial were compared with those of a clinical trial in which both imaging methods were used on 400 women, findings correlated well, confirming the superiority of digital breast tomosynthesis. An in-silico trial of an intervention (bone morphogenetic protein) in congenital pseudo-arthritis of the

tibia (a rare disease that affects fewer than 200 000 people in the USA) has been reported using 200 virtual patients.²⁷ This study supports using in-silico trials as an alternative in paediatric rare diseases for which patient clinical trials would be almost impossible to perform.²⁷ In-silico trials should be viewed as complementary to clinical trials when evaluating the performance of medical devices or interventions, and are cost-effective.

Advantages of virtual patient data

Synthetic control arms have expedited the evaluation of cancer medications,^{28,29} and interventions in hepatitis C and rheumatoid arthritis.³⁰ The pharmaceutical manufacturer AstraZeneca has advanced its digital clinical trials and has used over 300 million synthetic patient records, decreasing drug development costs (by an estimated US\$100 million per drug) and time.³¹ Commercial support to optimise clinical trials using virtual patients is currently available.³²

The use of virtual patient data in clinical trials—namely, digital twins, synthetic data, and in-silico trials—is emerging but has not spread to paediatric clinical trials. However, we would speculate that some of the completed trials would have benefited from using virtual patient data by improving power to detect a significant difference in outcomes between the comparison and intervention groups (eg, probiotic trials).⁴ Further research comparing controls used in real-world clinical trials and generated virtual patient data for control arms (synthetic or digital twin data) is necessary to confirm how accurate virtual patient data can be in terms of patient characteristics, which will increase our confidence in using virtual patient data for future paediatric clinical trials.

The use of virtual patient data to enhance paediatric clinical trials could have numerous advantages, including decreased exposure of children to potentially ineffective or risky interventions, shorter trial durations leading to more rapid ascertainment of safety and effectiveness of interventions, and faster drug approvals. Leveraging virtual patient data could ensure high certainty of evidence for interventions in children with more personalised treatment options. Incorporation of virtual patient data in clinical trials could lower costs associated with participant recruitment and management. Rapid ascertainment of safety and efficacy of interventions could shorten the duration of trials and benefit researchers, policy makers, and funding agencies. Researchers could have improved ability to test multiple treatment variations and test the potential for continuous monitoring and adaptive trial designs. We believe that the advantages of utilising virtual patient data would lead to enhanced data quality and quantity.

Regulations for the use of digital data

In the USA, the Food and Drug Administration (FDA) regulates and approves digital health technologies for clinical use, and digital health is considered a medical device. Digital health is regulated based on its intended use in the diagnosis, treatment, cure, mitigation, or prevention of disease or condition. The FDA has enforcement discretion,

which means that it does not comprehensively regulate low-risk medical devices or technologies. The FDA has implemented a Digital Health Center of Excellence,³³ whose goal is to empower stakeholders to advance health care by fostering responsible and high-quality digital health innovation, including building partnerships, increasing awareness, advancing best practices, and innovating regulatory approaches to provide efficient and the least burdensome supervision. The FDA advocates for in-silico modelling and simulation and acknowledges the potential of computational modelling to enhance the regulatory assessment process.³³ The FDA has also issued guidance on the use of computational modelling in medical device development.³⁴ In Europe, the Artificial Intelligence Act was passed by the EU parliament on March 13, 2024, and the law came into force on Aug 1, 2024.³⁵ The new law is applicable to the developers and deployers of general-purpose AI systems, to AI systems including AI enabled digital health tools, and to high-risk AI systems, including digital health tools in a staggered timeline. The Artificial Intelligence Act is a legislation with a broad scope compared to single-sector legislation (vertical legislation), which, for the health-care domain, are principally the Medical Device Regulation and In-Vitro Diagnostic Medical Devices Regulation, both specific to the EU. These two regulations will continue to apply to health-care AI alongside the Artificial Intelligence Act.³⁶ Effective and safe innovations in the use of virtual patient data for clinical trials require active collaboration and partnership of AI and computational experts, researchers, health-care providers, health-care institutions, and regulatory bodies.

Limitations, ethical concerns, and potential solutions

Digital twins and synthetic data for clinical trials and in-silico trials are in the early phase of adoption and several concerns must be addressed before widespread use. Validating the accuracy of synthetic data to represent real patient populations poses an obstacle, but this validation is crucial, particularly in drug discovery trials and research. Ensuring that synthetic data accurately reflect real-world diversity is essential to avoid biased or inconclusive outcomes. A major limitation of using virtual patient data in clinical trials is the inability to predict adverse events that have not been reported in previous datasets. Human biology is complex and all the interactions and outcomes of a new molecule or an intervention are impossible to predict; thus, serious adverse effects could be missed with virtual patient data. The placebo effect in patient clinical trials is real and leads to improved patient outcomes even in the control arm due to awareness of the health benefits of a study or better monitoring.^{37–39} Despite the advantages of in-silico trials or trials using synthetic data, a potential limitation is the lack of a placebo effect in the control arm. Furthermore, new ethical frameworks based on important ethical concepts of non-maleficence, justice, accountability, and respect for autonomy (including privacy) should be formulated.

Children are a vulnerable population typically represented by their parents or guardians. During clinical trials, they lack their own consent and voice. Meanwhile, parents or guardians might hesitate to enrol their children in clinical trials due to concerns about potential data leaks and subsequent ethical issues.⁴⁰ Researchers will have to embrace new governance and regulation regarding patient data privacy and data ownership to ensure the safety and sustainability of digital data innovation. Once implemented, strategies for managing costs and reimbursements should be established. Additionally, defining the relationship between virtual patient data (digital twin and synthetic) and clinicians is paramount as the risk of dehumanisation is a substantial concern. Reduced face-to-face interactions in clinical settings can diminish the level of trust between patients and health-care professionals, potentially leading to negative experiences and outcomes.⁴¹

A major risk is maintaining patient data privacy and preventing security breaches, which can be mitigated with the implementation of robust data protection measures and anonymisation techniques, and privacy by design environments for large-scale research data have been proposed.⁴² Patients need safeguards to protect their data from being exploited by others including insurers or employers. Digital data, although de-identified, should be protected from commercialisation. Whether patients can control the dissemination and utilisation of their digital health data, including digital twins, is unclear.⁴³ The ethical and legal aspects of loss of privacy by data breaches have to be formalised. Another potential risk is non-compliance with regulatory bodies and their regulations (once these have been formulated by experts with adequate consensus), which could be addressed by early and ongoing engagement with regulatory bodies and awareness about the regulatory principles. In a study to identify the major ethical risks of the use of digital twins in health care, Huang and colleagues report a process-oriented ethical map at the stages of data collection, data management, data analysis, and information use associated with a personalised health-care service using digital twins.⁴⁴ The aforementioned ethical risks should be balanced with the potential benefits of reducing the likelihood of assignment of children and their families to the control intervention (virtual patients could be enrolled in the control arm) and speedy decisions on the effectiveness and safety of new investigational therapies for children. Another key challenge is that formal instructions do not exist regarding consent of children for digital health technologies related to virtual patients in clinical trials. The National Institutes of Health guidance to consent for digital health technologies includes wearable technology but not those related to AI,⁴⁵ and so this is another key challenge that must be overcome.

Conclusion

In summary, the use of digital twins, synthetic patient data generation, and in-silico trials has the potential to enhance the sample size for paediatric clinical trials, thereby

For more on the **Artificial Intelligence Act** see <https://artificialintelligenceact.eu/>

For more on the **implementation timeline** see <https://artificialintelligenceact.eu/implementation-timeline/>

decreasing costs, and enabling results to be obtained faster. However, in addition to addressing numerous technical challenges of these technologies, we must design appropriate ethical, regulatory, and safety frameworks, and develop clear data privacy and ownership regulations before virtual patient data can be used widely in clinical research.

Contributors

MP conceptualised this study and wrote the first draft of the manuscript. PSS, JH, LKY, NA, and JN critically reviewed and revised the manuscript. All authors approved the final version of the manuscript.

Declaration of interests

The authors declare no competing interests.

Acknowledgments

This work was supported by a grant awarded to MP by NIH (R01HD112886), on which JN is a co-investigator and was partially awarded the grant.

References

- Oakes LM. Sample size, statistical power, and false conclusions in infant looking-time research. *Infancy* 2017; **22**: 436–69.
- Tarnow-Mordi WO, Robledo K, Marschner I, Seidler L, Simes J. To guide future practice, perinatal trials should be much larger, simpler and less fragile with close to 100% ascertainment of mortality and other key outcomes. *Semin Perinatol* 2023; **47**: 151789.
- Alsaied A, Islam N, Thalib L. Global incidence of necrotizing enterocolitis: a systematic review and meta-analysis. *BMC Pediatr* 2020; **20**: 344.
- Sharif S, Meader N, Oddie SJ, Rojas-Reyes MX, McGuire W. Probiotics to prevent necrotising enterocolitis in very preterm or very low birth weight infants. *Cochrane Database Syst Rev* 2020; **10**: CD005496.
- Tarnow-Mordi W, Cruz M, Morris J. Design and conduct of a large obstetric or neonatal randomized controlled trial. *Semin Fetal Neonatal Med* 2015; **20**: 389–402.
- Schmidt B, Whyte RK. Oxygen saturation target ranges and alarm settings in the NICU: what have we learnt from the neonatal oxygenation prospective meta-analysis (NeOProm)? *Semin Fetal Neonatal Med* 2020; **25**: 101080.
- Collaboration A. Alpha Collaboration: Improving perinatal health through large, simple trials. 2024. <https://www.alphacollaboration.com> (accessed Oct 7, 2024).
- National Academies of Sciences, Engineering, and Medicine. Foundational Research Gaps and Future Directions for Digital Twins. March 28, 2024. <https://nap.nationalacademies.org/catalog/26894/foundational-research-gaps-and-future-directions-for-digital-twins> (accessed April 2, 2025).
- Drummond D, Gonsard A. Definitions and characteristics of patient digital twins being developed for clinical use: scoping review. *J Med Internet Res* 2024; **26**: e58504.
- Venkatesh KP, Raza MM, Kvedar JC. Health digital twins as tools for precision medicine: considerations for computation, implementation, and regulation. *NPJ Digit Med* 2022; **5**: 150.
- Katsoulakis E, Wang Q, Wu H, et al. Digital twins for health: a scoping review. *NPJ Digit Med* 2024; **7**: 77.
- Cellina M, Cè M, Ali M, et al. Digital twins: the new frontier for personalized medicine? *Appl Sci* 2023; **13**: 7940.
- Stevenson DK, Wong RJ, Aghaepour N, et al. Towards personalized medicine in maternal and child health: integrating biologic and social determinants. *Pediatr Res* 2021; **89**: 252–58.
- Calcaterra V, Pagani V, Zuccotti G. Digital twin: a future health challenge in prevention, early diagnosis and personalisation of medical care in paediatrics. *Int J Environ Res Public Health* 2023; **20**: 2181.
- Gonsard A, AbouTaam R, Prévost B, et al. Children's views on artificial intelligence and digital twins for the daily management of their asthma: a mixed-method study. *Eur J Pediatr* 2023; **182**: 877–88.
- Sizemore N, Oliphant K, Zheng R, Martin CR, Claud EC, Chattopadhyay I. A digital twin of the infant microbiome to predict neurodevelopmental deficits. *Sci Adv* 2024; **10**: ead0400.
- Wang Y, Fu T, Xu Y, et al. TWIN-GPT: digital twins for clinical trials via large language model. *ACM Trans Multimedia Comput Commun Appl* 2024; published online July 10. <https://doi.org/10.1145/3674838>.
- Rajotte JF, Bergen R, Buckeridge DL, El Emam K, Ng R, Strome E. Synthetic data as an enabler for machine learning applications in medicine. *iScience* 2022; **25**: 105331.
- Chen RJ, Lu MY, Chen TY, Williamson DFK, Mahmood F. Synthetic data in machine learning for medicine and healthcare. *Nat Biomed Eng* 2021; **5**: 493–97.
- Giuffrè M, Shung DL. Harnessing the power of synthetic data in healthcare: innovation, application, and privacy. *NPJ Digit Med* 2023; **6**: 186.
- Jordon J, Szpruch L, Houssiau F, et al. Synthetic data—what, why and how? *arXiv* 2022; published online May 6. <https://doi.org/10.48550/arXiv.2205.03257> (preprint).
- Kokosi T, Harron K. Synthetic data in medical research. *BMJ Med* 2022; **1**: e000167.
- Bertolini D, Loukianov AD, Smith A, et al. Forecasting progression of mild cognitive impairment (MCI) and Alzheimer's disease (AD) with digital twins. *Alzheimers Dement* 2021; **17** (suppl 9): e054414.
- D'Amico S, Dall'Olio D, Sala C, et al. Synthetic data generation by artificial intelligence to accelerate research and precision medicine in hematology. *JCO Clin Cancer Inform* 2023; **7**: e2300021.
- Kolla L, Gruber FK, Khalid O, Hill C, Parikh RB. The case for AI-driven cancer clinical trials - the efficacy arm in silico. *Biochim Biophys Acta Rev Cancer* 2021; **1876**: 188572.
- Badano A, Graff CG, Badal A, et al. Evaluation of digital breast tomosynthesis as replacement of full-field digital mammography using an in silico imaging trial. *JAMA Netw Open* 2018; **1**: e185474.
- Carlier A, Vasilevich A, Marechal M, de Boer J, Geris L. In silico clinical trials for pediatric orphan diseases. *Sci Rep* 2018; **8**: 2465.
- Wedam S, Fashoyin-Aje L, Bloomquist E, et al. FDA approval summary: palbociclib for male patients with metastatic breast cancer. *Clin Cancer Res* 2020; **26**: 1208–12.
- Davies J, Martinec M, Delmar P, et al. Comparative effectiveness from a single-arm trial and real-world data: alectinib versus ceritinib. *J Comp Eff Res* 2018; **7**: 855–65.
- Thorlund K, Dron L, Park JJH, Mills EJ. Synthetic and external controls in clinical trials - a primer for researchers. *Clin Epidemiol* 2020; **12**: 457–67.
- Durán C, Bonam M. Clinical Innovation: how digital health solutions are transforming our trials. Sept 26, 2023. <https://www.astrazeneca.com/what-science-can-do/topics/clinical-innovation/clinical-innovation-digital-health-solutions-transforming-trials.html> (accessed Oct 7, 2024).
- Unlearn AI. Trial design in your hands. 2024. <https://www.unlearn.ai/> (accessed April 2, 2025).
- US Food and Drug Administration. Digital Health Center of Excellence. 2024. <https://www.fda.gov/medical-devices/digital-health-center-excellence> (accessed Oct 7, 2024).
- US Food and Drug Administration. CFDAH. Assessing the credibility of computational modeling and simulation in medical device submissions. Docket number: FDA-2021-D-0980. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-device-submissions> (accessed April 2, 2025).
- Gilbert S. The EU passes the AI Act and its implications for digital medicine are unclear. *NPJ Digit Med* 2024; **7**: 135.
- Gilbert S, Anderson S, Daumer M, Li P, Melvin T, Williams R. Learning from experience and finding the right balance in the governance of artificial intelligence and digital health technologies. *J Med Internet Res* 2023; **25**: e43682.
- Enck P, Klosterhalfen S. Placebos and the placebo effect in drug trials. *Handb Exp Pharmacol* 2019; **260**: 399–431.
- Enck P, Klosterhalfen S. The placebo response in clinical trials—the current state of play. *Complement Ther Med* 2013; **21**: 98–101.
- Wampold BE, Minami T, Tierney SC, Baskin TW, Bhati KS. The placebo is powerful: estimating placebo effects in medicine and psychotherapy from randomized clinical trials. *J Clin Psychol* 2005; **61**: 835–54.

- 40 Braun M, Krutzinna J. Digital twins and the ethics of health decision-making concerning children. *Patterns (N Y)* 2022; **3**: 100469.
- 41 Drummond D, Coulet A. Technical, ethical, legal, and societal challenges with digital twin systems for the management of chronic diseases in children and young people. *J Med Internet Res* 2022; **24**: e39698.
- 42 Zhang P, Kamel Boulos MN. Privacy-by-design environments for large-scale health research and federated learning from data. *Int J Environ Res Public Health* 2022; **19**: 11876.
- 43 Venkatesh KP, Brito G, Kamel Boulos MN. Health digital twins in life science and health care innovation. *Annu Rev Pharmacol Toxicol* 2024; **64**: 159–70.
- 44 Huang PH, Kim KH, Schermer M. Ethical issues of digital twins for personalized health care service: preliminary mapping study. *J Med Internet Res* 2022; **24**: e33081.
- 45 NIH Office of Science Policy. Informed consent for research using digital health technologies: points to consider & sample language. May, 2024. https://osp.od.nih.gov/wp-content/uploads/2024/05/DigitalHealthResource_Final.pdf (accessed April 2, 2025).

© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).